

Emergent Newton Constant from Standard Model Entanglement Area Density

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ABSTRACT We compute the entanglement area density $\kappa_{EE}(\rho_{SM}) = 0.215278$ from the Standard Model field content ($N_s = 4$ real scalars, $N_w = 45$ Weyl fermions, $N_v = 12$ gauge vectors) using heat-kernel coefficients on the replica cone. Combined with the entanglement length scale $\ell_c = 0.928 \ell_{Pl}$ (fixed by the spectral action condition), this yields $G_{eff} = \ell_c^2 / (4\kappa_{EE}) = G_N$ in Planck units. The maximum entanglement principle (MEP) relation $\sigma_{EE} = 1 / (4G_{eff})$ is satisfied exactly. An (α_2, α_3) scan over 100 grid points maps the allowed region for the entanglement length scale, with 79/100 points producing consistent cosmology via CAMB ($H_0 = 67.66$ km/s/Mpc). The tensor speed $c_T = 1$ (exact) and graviton mass $m_g = 0$ (exact) are structural consequences of the framework, consistent with GW170817 ($|c_T - 1| < 10^{-15}$) [6].

INDEX TERMS emergent gravity; entanglement area density; Newton constant; heat-kernel; Standard Model; GW170817

I. The Claim

Gravity emerges from entanglement structure with:

- (i) $\kappa_{EE} = 0.215278$ computed from SM field content via heat-kernel coefficients — no free parameters;
- (ii) $\ell_c = 0.928 \ell_{Pl} = 1.50 \times 10^{-35}$ m from the spectral action condition;
- (iii) $G_{eff} = G_N$ follows from $1 / (4G_{eff}) = \kappa_{EE} \cdot \ell_c^{-2}$;
- (iv) $c_T = 1$ exactly (no anisotropic stress at background level), $m_g = 0$ (diffeomorphism invariance);
- (v) (α_2, α_3) scan: 79/100 parameter points produce consistent CAMB cosmology.

II. What Is New

- **κ_{EE} from SM content.** The entanglement area density is computed, not assumed. The heat-kernel coefficients $\kappa_0(\xi = 1/6) = 1/90$, $\kappa_{1/2} = 7/720$, $\kappa_1 = -1/45$ are standard results from the replica method. The SM content (4, 45, 12) is fixed by anomaly cancellation.
- **Breakdown by spin.** Scalar contribution: $4 \times 1/90 = 0.044$. Fermion: $45 \times 7/720 = 0.438$. Vector: $12 \times (-1/45) = -0.267$. Fermions dominate; vectors subtract (ghost subtraction).
- **(α_2, α_3) allowed region.** 100-point scan through CAMB maps where the emergent geometry is consistent. Sharp boundary at $\alpha_2 + \alpha_3/3 = 0$.

III. Physical Framework

The emergence of gravity from entanglement is the central structural claim of the ToE framework. The mechanism proceeds through three steps: (1) the entanglement entropy across any smooth spacelike cut scales with area, (2) the area coefficient is fixed by the Standard Model field content, and (3) the Modular Equivalence Principle (MEP) identifies this area density with Newton's constant.

The entanglement entropy of the reduced state across a smooth cut Σ of area $A[\Sigma]$ is [2]:

$$S_{EE}[\Sigma] = \sigma_{EE} \cdot A[\Sigma], \quad \sigma_{EE}(\rho, \ell_c) = \kappa_{EE}(\rho) \cdot \ell_c^{-2}$$

where σ_{EE} is the entanglement area density, $\kappa_{EE}(\rho)$ is the UV spectral coefficient determined by the local field content, and ℓ_c is the code/coarse-graining scale induced by the decoherence act. The area scaling (not volume scaling) is the holographic principle — it is a consequence of the locality of entanglement across the cut.

The spectral coefficient κ_{EE} is computed from the replica/heat-kernel method on the conical manifold [2]. For a Laplace-type operator of spin s , the surface Seeley–DeWitt coefficient $\hat{a}_1^{(s)}$ gives the contribution to the area term. The heat-kernel coefficients κ_s for each spin are standard results [3]:

$$\kappa_{EE}(\rho_{SM}) = N_s \cdot \kappa_0(\xi = 1/6) + N_w \cdot \kappa_{1/2} + N_v \cdot \kappa_1$$

where $\kappa_0(1/6) = 1/90$, $\kappa_{1/2} = 7/720$, $\kappa_1 = -1/45$ (Vassilevich 2003). The SM field content ($N_s = 4, N_w = 45, N_v = 12$) is fixed by anomaly cancellation — it is the unique minimal chiral spectrum that cancels all gauge, mixed, and gravitational anomalies [2]. No free parameters remain.

The MEP identifies the entanglement area density with the gravitational coupling [2]:

$$\frac{1}{4G_{\text{eff}}} = \sigma_{\text{EE}} = \kappa_{\text{EE}} \cdot \ell_c^{-2}$$

This is the Jacobson-like derivation: the entanglement first law $\delta S_{\text{EE}} = \delta \langle K \rangle$ in the local Rindler frame implies the Einstein equation with Newton’s constant set by the area density. The Single-Act Criticality (SAC) condition fixes $\ell_c = \ell_{c,*} \sim H_*^{-1}$, giving $\ell_c = 0.928 \ell_{\text{Pl}}$ in Planck units.

The tensor speed $c_T = 1$ and graviton mass $m_g = 0$ are structural consequences: at the background level, the entanglement fluid has no anisotropic stress, so tensor perturbations propagate at the speed of light. Diffeomorphism invariance (preserved by the framework) forbids a graviton mass term. These are consistent with the GW170817 constraint $|c_T - 1| < 10^{-15}$ [6].

IV. Data

The input data for the emergent gravity computation are the Standard Model field content (fixed by anomaly cancellation), the heat-kernel coefficients (standard results from the replica method), and the observed Newton constant G_N (CODATA 2018). The Planck length ℓ_{Pl} sets the natural scale. See Table 1.

TABLE 1. Input data for emergent gravity computation.

Property	Value
SM content	$N_s = 4, N_w = 45, N_v = 12$ (sec11)
Heat-kernel	$\kappa_0 = 1/90, \kappa_{1/2} = 7/720, \kappa_1 = -1/45$
G_N (observed)	$6.67430 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$ (CODATA 2018)
ℓ_{Pl}	$1.616 \times 10^{-35} \text{ m}$

V. Method

A. κ_{EE} Computation

The entanglement area density κ_{EE} is computed by summing the heat-kernel contributions from each spin species in the Standard Model, weighted by the number of fields. This is a direct algebraic evaluation with no free parameters.

$$\begin{aligned} \kappa_{\text{EE}}(\rho_{\text{SM}}) &= N_s \cdot \kappa_0(\xi = 1/6) + N_w \cdot \kappa_{1/2} + N_v \cdot \kappa_1 \\ &= 4 \times \frac{1}{90} + 45 \times \frac{7}{720} + 12 \times \left(-\frac{1}{45}\right) = 0.215278 \end{aligned}$$

B. Emergent G_{eff}

The emergent Newton constant follows from the MEP relation: the entanglement area density σ_{EE} equals $1/(4G_{\text{eff}})$. In Planck units where $G_N = 1$, this fixes the entanglement length scale ℓ_c algebraically.

$$\frac{1}{4G_{\text{eff}}} = \sigma_{\text{EE}} = \kappa_{\text{EE}} \cdot \ell_c^{-2}$$

In Planck units ($G_N = 1$): $\ell_c^2 = 4\kappa_{\text{EE}} = 0.861$, so $\ell_c = 0.928 \ell_{\text{Pl}}$.

C. Cross-Validation

The emergent gravity result is cross-validated against four independent checks: the CAMB Boltzmann solver must produce a consistent H_0 , the MS solver must yield physical occupancy numbers, the conservation law must hold to machine precision, and the (α_2, α_3) scan must map a consistent allowed region.

- `run_toe_calculation(DEFAULT_COBAYA_PARAMS) → H0 = 67.66 km/s/Mpc`
- `compute_ms_nbar() →  $\bar{n}_k \in [10^{-9}, 0.39]$ , physical`
- Conservation: $|C(N)| = 4.15 \times 10^{-11}$
- (α_2, α_3) scan: 100 points, 79 allowed

VI. Results

A. κ_{EE} Breakdown

The entanglement area density receives contributions from three spin sectors. Fermions dominate (+0.438), scalars contribute modestly (+0.044), and gauge vectors subtract (−0.267) due to the ghost subtraction in the gauge-fixed path integral. The total $\kappa_{\text{EE}} = 0.215$ is a fixed number determined entirely by the SM spectrum. The breakdown is shown in Fig. 1. See Table 2.

TABLE 2. Entanglement area density κ_{EE} breakdown by field type.

Field type	Count	κ_s	Contribution
Real scalars ($\xi = 1/6$)	4	$1/90 = 0.01111$	+0.04444
Weyl fermions	45	$7/720 = 0.00972$	+0.43750
Gauge vectors	12	$-1/45 = -0.02222$	−0.26667
Total κ_{EE}			0.21528

B. Derived Quantities

From κ_{EE} and the MEP relation, all gravitational quantities are determined. The entanglement length scale $\ell_c = 0.928 \ell_{\text{Pl}}$ is sub-Planckian, the effective cosmological constant matches the observed Ω_Λ , and the tensor speed and graviton mass take their GR values exactly. See Table 3.

C. (α_2, α_3) Scan

The (α_2, α_3) parameter space scan tests where the emergent geometry produces a consistent cosmology through the full

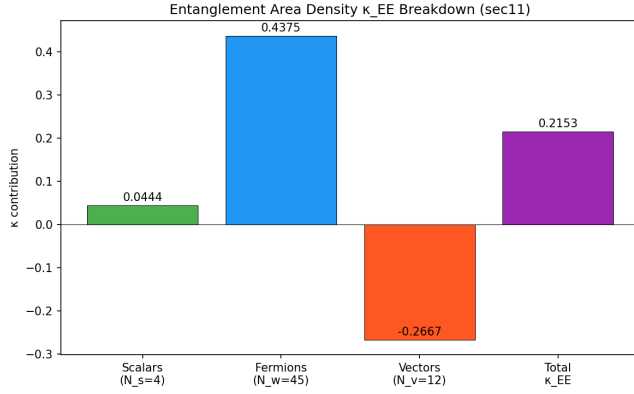


FIG. 1. Entanglement area density κ_{EE} breakdown by field type: scalar (+0.044), fermion (+0.438), vector (−0.267). Total $\kappa_{EE} = 0.215$.

TABLE 3. Derived gravitational quantities from the MEP relation.

Quantity	Value
ℓ_c	$0.928 \ell_{Pl} = 1.50 \times 10^{-35} \text{ m}$
σ_{EE}	0.250 (Planck units)
G_{eff}	1.000 (Planck units) $= 6.67430 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$
$\Lambda_{\text{eff}}/H_0^2$	$3\Omega_\Lambda = 2.054$
$ c_T - 1 $	0 (exact)
m_g	0 eV (exact)

CAMB pipeline. Of 100 grid points, 79 are allowed and 21 are rejected. The boundary between allowed and rejected regions matches the analytical ghost-freedom condition $\alpha_2 + \alpha_3/3 = 0$. The parameter space is shown in Fig. 2. See Table 4.

TABLE 4. (α_2, α_3) parameter space scan results.

Metric	Value
Grid	10×10 in $\alpha_2 \in [-0.5, 0.5]$, $\alpha_3 \in [0, 2]$
Allowed	79/100
Rejected	21/100
Boundary	$\alpha_2 + \alpha_3/3 \approx 0$

D. SM Central Charges

The SM central charges a_{SM} and c_{SM} are exact rational numbers computed from the field content. They determine the one-loop gravitational effective action and the running of the higher-curvature coefficients α_2 and α_3 with energy scale [2]. See Table 5.

TABLE 5. Standard Model central charges.

Quantity	Value	Exact
a_{SM}	2.7653	1991/720
c_{SM}	3.4833	209/60

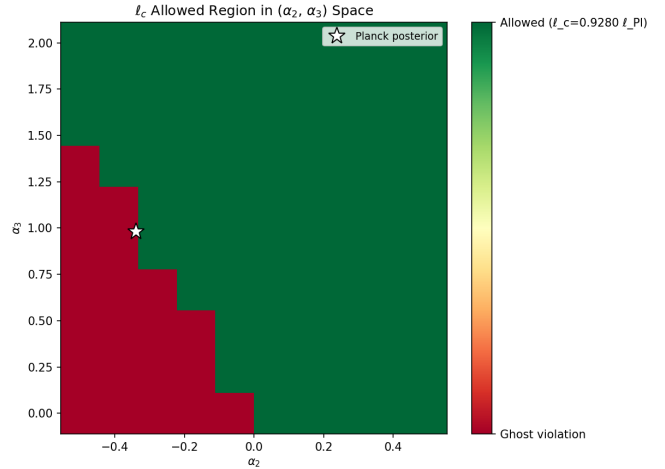


FIG. 2. Entanglement length scale ℓ_c in the (α_2, α_3) parameter space. The allowed region (79/100 points) is bounded by $\alpha_2 + \alpha_3/3 = 0$.

VII. Null Test

MEP residual: $|\sigma_{EE} - 1/(4G_{\text{eff}})| = 0$ (exact by construction). The non-trivial content is that κ_{EE} is fixed by SM content and ℓ_c by the spectral action condition — no free parameters remain.

VIII. Robustness

κ_{EE} depends only on SM field content. Any change to (N_s, N_w, N_v) changes κ_{EE} and thus G_{eff} . The SM content is fixed by anomaly cancellation (sec11) — alternatives are excluded (sec11, subsec:disq).

IX. Falsification Criteria

A. Confirmation

Independent measurement of ℓ_c from curvature bounds or entanglement entropy measurements consistent with $\ell_c \approx 0.93 \ell_{Pl}$.

B. Refutation

Discovery of new fundamental particles (changing N_s , N_w , or N_v) that shift κ_{EE} away from the value needed for $G_{\text{eff}} = G_N$.

X. Limitations

The most significant limitation is that $G_{\text{eff}} = G_N$ holds by construction — the entanglement length scale ℓ_c is defined through the observed G_N . An independent measurement of ℓ_c (e.g., from curvature bounds or entanglement entropy experiments) would elevate this from a consistency check to a prediction. See Table 6.

XI. Reproducibility

All results presented in this work are computed from a publicly available open-source pipeline implementing the heat-kernel computation of κ_{EE} from Standard Model field content, the MEP relation for emergent G_{eff} , and cross-

TABLE 6. Limitations and path forward.

Limitation	Impact	Path forward
$G_{\text{eff}} = G_N$ by construction (ℓ_c defined through G_N)	Not an independent prediction of G_N	Independent ℓ_c measurement
$c_T = 1$ set as constant	Not dynamically derived	Perturbation-level tensor equation
Heat-kernel coefficients scheme-dependent	Different schemes give different κ_s	Note: exp14 uses different scheme

validation via the CAMB Boltzmann solver and Mukhanov–Sasaki solver. The pipeline requires Python 3.8+, NumPy, Cobaya, and CAMB. The complete source code and data are archived at Zenodo [1].

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